Yulong Ge

Ph.D. Student in Applied Mathematics

Harbin, China | geyvlong@stu.hit.edu.cn | yulong-ge.github.io | yulong-ge

Halongbay_AI

Education

Ph.D. **Harbin Institute of Technology**, Applied Mathematics — Harbin, China
- Supervised by Professor Jianwei Ma.
- Research focuses on computer vision, generative models, and neural network interpretability.

B.S. **Harbin Institute of Technology**, Information and Computing Science — Harbin, China

Minor **Harbin Institute of Technology**, Big Data Management and Applications — Harbin, China

Research and Project Experience

Domain-Adaptive Deep Learning Feature Space Research

Collaborative research on feature-space structure to improve deep-model robustness under distribution shifts.
- Studied structural properties of model feature spaces for generalization to unseen domains.
- Related research outcome published in IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI).

Intelligent Identification and Analysis of Martian Geological Faults

AI for Science project for identifying geological faults in Martian data.
- Developed core components of an AI fault-recognition pipeline.
- Vectorized and standardized fault data for downstream analysis.

Efficient Sampling Algorithms for Diffusion Generative Models

Undergraduate thesis on improving the sampling efficiency of diffusion models.
- Investigated and designed algorithms targeting a central efficiency challenge in diffusion-model sampling.

Generative-Model-Based Ultrasound Medical Image Translation

Cross-modal medical-image translation research using generative models.
- Explored generation of diagnostically informative images to address limitations of single-modality ultrasound data.

Open Source

AI Feed Tracker

AI-subscription service that summarizes time-sensitive updates close to real time.

OpenChamber

Open-source project contributor.

Technical Skills

Mathematics and Theory: Computational mathematics, applied mathematics, linear algebra, probability, statistical learning theory, variational inference

Large Language Models: Foundations of LLM fine-tuning and practical LoRA fine-tuning

Generative Modeling: Diffusion models for natural-image generation

Embedded Systems: STM32 development and SolidWorks mechanical modeling

Languages: Chinese (native); English (CET-6 579, CET-4 593)

Honors and Awards

Awards: People's Scholarship; Outstanding Youth League Member; Outstanding Student; Special Graduate Academic Scholarship

Volunteer Experience

Computer 110 Club — 2020-2022

Supported on-campus computer diagnostics and wrote new-semester purchasing guides for students.